

Ridham Bhagat

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EDUCATION

- **Master of Science in Cybersecurity, Northeastern University** Sept 2024 – May 2026
Coursework: Computer Systems Security, Network Security, Software Vulnerabilities and Security, Distributed Systems, Embedded Software development and Security
GPA: 3.917
- **Bachelor of Technology in Computer Science, Shiv Nadar University** May 2019 – May 2023

TECHNICAL SKILLS

- **Programming Languages:** Java, JavaScript, Python, C, C#, x86 Assembly, Bash, Solidity, Go
- **Cybersecurity:** Network Penetration Testing, DevSecOps, Secure Code Review, Binary Exploitation
- **Tools:** BurpSuite, Jadx, Nmap, IPtables, Wireshark, Ghidra, GDB, Pwntools, Splunk, Wazuh, Active Directory, Sysinternals
- **Cloud and DevOps:** AWS, GCP, Docker, Kubernetes, Terraform, Digital Ocean, Run Pod, Linux, CDN, Ansible
- **Software & Data Technologies:** Spring Boot, NodeJS, MySQL, MongoDB, Graph QL, Angular

WORK EXPERIENCE

Security Engineer , DevKit April 2022 – June 2023

- Implemented **rate-limiting** using Uptash, a serverless Redis provider, to prevent abuse of AI endpoints
- Implemented **JWT authorisation** to secure all **10+ backend endpoints**, updated **5+ dependencies** to prevent supply chain attacks
- Added **guardrails** to defend against basic **prompt injection attacks**.
- Concurrently, served as an **incident responder** and handled instances of security breaches due to previously misconfigured systems.

Software Engineer , Dell Technologies Aug 2023 – Feb 2024

- Developed a **utility tool to query Elasticsearch** for logs, performing SQL-like joins and filters on multiple indices, utilising **hashmaps, Roaring bitmaps and GraphQL**. Built the MVP capable of processing **5 million log** entries per run.
- **Revamped old legacy backend** systems and migrated them to **Kubernetes**
- Integrated **DevSecOps** into the new applications pipeline, resulting in prioritisation of the **resolution of security holes** like vulnerable dependencies in the applications. The overall impact was a **35% reduction in time-to-detect** post-integration.
- Implemented a better **logging mechanism**, resulting in ease in monitoring, leading to **faster triaging of production issues**.

Process Engineer , Addverb Technologies May 2025 – Sept 2025

- Deployed Fleet management system to AWS and managed on-prem communication latencies with robots.
- **Stabilised networks** using 5 GHz channel/TX-power tuning , allows for easier roaming of clients.
- Created **custom firmware** for robot's network utilities to enable easier roam.
- Built **hourly/robot/AP heatmaps** and dashboards to visualize **RSSI/SNR**, retries, and roam timing next to robot incidents.
- Built ops tooling involving Ansible playbooks + FastAPI Backend + Angular UI; Docker + K8s deploys behind NGINX. Managed fleet of **250+ Robots**.

Research Assistant, Northeastern University Jan 2026 - April 2026

- Working on analysing global internet topology to create AS+IXP multigraphs
- Utilising the AS+IXP multigraphs to understand eBGP connections for proxy placements to circumvent state-implemented censorship using Refraction Networking.

ACHIEVEMENTS

- Represented Northeastern University at the **MITRE embedded CTF competition**. Ranked 16th.
- Won **1st prize** in **NCAE CyberGames Regionals** among **26 teams**.
- Won **3rd place** in **CCDC Invitationals** among **21 teams**, a prestigious blue-team cybersecurity competition.
- Represented Northeastern University in **DoE Cyberforce Competition**. specializing in Windows security, logging, and network infrastructure detection.
- Currently on the competing roster for the **CCDC competition** at Northeastern University. Handling network infra

PROJECTS

- **DNS Monitoring tooling** - Currently developing a **DNS monitoring tool** for **detecting supply chain attacks** in web3 projects in collaboration with [Digibastion](#). Received a grant from the Ethereum Foundation.
- **Binary Exploitation** - Automated format-string exploits with Pwntools; built advanced **ROP chains** (libc + syscalls), used **ret2plt** with **.got** leaks to **bypass ASLR**, plus **stack pivoting** and **canary bypass** to achieve code execution.
- **Custom Messenger Protocol** [\[Link\]](#) - Designed and implemented a secure **cryptographic messenger protocol** by using existing cryptographic primitives and carried out its **formal verification** in **ProVerif**.
- **QOS AP** [\[Link\]](#) - Created a plug-and-play AP with custom QOS rules to reduce latency while playing online games.